Score Conditioning and Static Threshold Calibration in Multivariate Time-Series Anomaly Detection: An Empirical Study

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***Abstract*—Automated anomaly thresholds can generate excessive alarms when telemetry scores change between calibration and deployment. We compare latent predictive, observation-space, and simple bounded scores under chronological held-out calibration on 33 streams from seven datasets; a 19-stream non-GHL cohort supports the five-seed controls. On that cohort, the latent predictor has lower nominal false-positive rates than two observation-space reconstruction baselines, but a bounded, dimensionality-matched principal-component control and a bounded observation-space TCN control also produce light residual tails. The TCN control nevertheless has higher calibration false-positive rates, showing that tail shape and threshold transfer are distinct. Causal event-recall curves suggest a different ordering when plotted against realized false-positive rate, but paired interpolation on a common observed range covers few streams and does not establish an aggregate advantage; buffered attribution is analyzed separately. Point-F1 and precision-recall differences remain imprecise, not equivalent. Static peaks-over-threshold calibration substantially overshoots its target risk under this short-block protocol. Inter-block score drift and scarce tail exceedances are both plausible contributors; their relative causal roles remain unresolved. Three physics-inspired variants show no reliable aggregate detection gain. These findings emphasize matched score controls, reporting of realized alarm rates, and limits on what nominal-budget curves can establish.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In remote sensing deployments — NASA SMAP, MSL rovers, industrial IoT facilities, wearable biometric arrays — dozens to hundreds of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3], and faults must be identified before hardware damage occurs.

A. Challenges in Modeling Multi-Sensor Telemetry

Multi-sensor telemetry reflects a physical state trajectory $\dot{x}(t) = \mathbf{f}(x(t))$, yet modeling dynamics in raw observation space faces four obstacles: estimating derivatives $\dot{x} \approx \Delta x/\Delta t$ on noisy discrete streams amplifies high-frequency noise; telemetry captures only a noisy projection of unmeasured states, violating full-state observability; in large arrays ($K \geq 25$) cross-channel nonlinear interactions resist rigid functional forms; and discrete operational mode transitions (orbital day/night cycling, thruster firings, gait freezing) induce continuous distribution drift.

B. Limitations of Deep Anomaly Detectors

Reconstructive models score observation-space errors, whereas latent predictors and low-dimensional controls use differently scaled, often bounded scores [3]–[5]. Comparing their alarm rates at a common nominal threshold risk does not separate scoring geometry from discriminative quality. Nor does a static threshold fitted on one nominal block guarantee the same false-positive rate on a later block. Whether high-frequency noise specifically consumes reconstruction capacity, or synthetic perturbations in contrastive methods such as NCAD [6] distort decisions, remains untested here.

C. Investigating Latent Prediction and Inductive Priors

We compare score conditioning, detection, and threshold transfer separately. A JEPA predicts target representations from context [7], [8]; observation-space models, a matched-backbone control, a LayerNorm-bounded observation-space TCN, and a 32-dimensional PCA forecast with normalized gated scoring test alternative explanations. We also evaluate spectral-entropy, latent second-moment, and potential-flow variants as exploratory physics-inspired controls. These experiments do not by themselves isolate a noise-removal mechanism.

D. Contributions

This work makes four principal contributions:

1) Disentangling Alarm Rates and Score Quality: On 19 non-GHL streams the nominal held-out false-positive rate is 7.6% for TS-JEPA versus 30% for TimesNet, but bounded PCA and bounded observation-space TCN controls also produce light residual tails, and the TCN control does not preserve

low calibration FPR. The larger JEPA event-recall area over *nominal design budgets* under causal attribution does not establish dominance at matched realized FPR: few streams span the same realized-FPR range, and paired comparisons are inconclusive. Buffered and causal results are reported separately.

2) Null Results for Physics-Inspired Variants: Across the 33-stream SPOT cohort, the tested regularizers provide no reliable aggregate detection gain. A Potential-Flow Point-F1 difference has an unadjusted p=0.033 but does not justify a multiplicity-adjusted superiority or inferiority claim.

3) Conditional Calibration Failure: Static POT thresholds substantially exceed their nominal design risk under short held-out calibration. Score-distribution differences across blocks correlate with false alarms, but small effective exceedance counts and observational confounding prevent causal attribution to drift alone.

4) Reproducible Multi-Seed Benchmark: The SPOT benchmark spans 33 streams from seven datasets (19 non-GHL streams with five seeds; 14 GHL streams with fewer seeds). The score-dump operating-curve cohort is narrower: 19 buffered and 18 causal non-GHL streams. We report paired seed coverage and dataset-level uncertainty for the matched-realized-FPR analysis.

Supplementary material contains proofs, extended diagnostics, and case-study detail.

II. Related Work

A. Dynamical Systems, Reconstruction, and Forecasting Detectors

Physical modeling of multi-sensor telemetry frames system behavior as continuous trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [9]. Neural ODEs [10] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness and jitter sensitivity. Koopman theory and DMD [11]–[13] instead learn linear operators on observable embeddings; physical disruptions then manifest as spectral decoherence. Our Operator-Entropy variant transfers this idea to a latent transition operator regularized by spectral entropy during training.

The predominant detection paradigm learns nominal patterns via raw-signal reconstruction: LSTM prediction with dynamic thresholding for NASA telemetry [3], association-discrepancy attention in Anomaly Transformer [4], 2D FFT-periodicity decomposition in TimesNet [5], and adversarial sub-series attention in TranAD [14]. These share a vulnerability: point-level losses bias networks toward fitting stochastic noise, and sustained anomalies contaminate the context window. These errors accumulate residual variance across irrelevant channels in high-dimensional arrays.

B. Threshold Calibration, Contrastive, and Self-Supervised Detection

Unsupervised detectors require an automated rule converting residual scores to binary alarms without test labels. Nonparametric dynamic thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [15]—fitting a Generalized Pareto tail above a high quantile under design risk q—are standard; however, EVT's asymptotic guarantees assume i.i.d. tail samples, which telemetry autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods learn discriminative representations: NCAD [6] trains against synthetic anomalies, yielding lower calibrated detection than predictive models (0.058 vs. 0.160 macro Point-F1, Supplementary Table S26). DCdetector [16] maximizes cross-scale dual-attention consistency. Joint-Embedding Predictive Architectures [7], [8], [17] instead predict target representations in latent space, with non-contrastive collapse prevention [18]–[20]. Time-series extensions [21] evaluate ranking or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration—the gap this work addresses.

C. Physics-Inspired Inductive Biases

PINNs [22] and Hamiltonian/Lagrangian networks [23], [24] embed dynamics as hard constraints; their stiff competing gradients [25] and closed-form-PDE requirements fit poorly to uncalibrated multi-sensor telemetry. We instead test whether *inspired* structures—spectral entropy, second-moment closure, conservative potential flow—improve latent residuals.

D. Terminology and Positioning

We distinguish *physics-inspired* structures (exploratory regularizers; latent coordinates are not claimed to be physical state variables) from *physics-informed* constraints enforcing known governing equations; *causal preprocessing* (features computed strictly from history) from *causal decision-time access* (alarm latency); and transition-operator non-degeneracy from representation anti-collapse. This investigation bridges joint-embedding predictive learning and multivariate anomaly detection: three physics-inspired regularizers are evaluated against unconstrained latent controls, and EVT calibration is evaluated on strictly held-out nominal telemetry to quantify the gap between design risk and realized false-alarm rates.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [26], [27], TimeEval [28], and mTSBench [29]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP Earth-observation `A-1`/`A-2`/`A-7`/`D-3`/`P-3`/`E-1` power, thermal, and attitude subsystems; MSL rover `C-1`/`C-2`/`D-14`/`D-15`/`D-16`/`M-1` actuator and environmental channels, with two short-nominal streams reserved; SWAN-SF solar magnetograms [30]); *wearable kinematics* (six Daphnet FoG streams [26]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so S=64 is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [31], six

CICIDS edge network-flow streams, CalIt2 people-flow [27], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [28] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample calibration, each stream's nominal sequence is partitioned chronologically into training (65%), calibration (17.5%), and held-out evaluation (17.5%) blocks separated by an isolation gap (Section V-A). The held-out protocol spans 33 streams across seven datasets (6 Daphnet, 14 GHL, 4 MSL, 6 SMAP, Genesis, Metro, SWAN); the remaining 12 streams are reserved as operational case studies — CalIt2, two Room-Occupancy, two short-nominal MSL, GECCO, and six CICIDS — because their nominal blocks are too short for disjoint partitions or because multi-seed stride-1 evaluation over 10^6 steps was prohibitive. The dyadic front-end ablation (Table Ib) uses a separate five-stream panel that additionally draws one SMD channel outside this manifest.

Two caveats bound interpretation. SMAP and MSL contain documented mislabeled segments and near-trivial anomalies [32]; their per-dataset numbers are upper bounds on detectability. Entity coverage is also incomplete — 6 of 55 SMAP and 4 of 27 MSL channels, selected by nominal-block length and stride-1 runtime constraints, not labels — so claims apply to the evaluated entities, not the full archives.

Supplementary Table S14 specifies the suite: 2,079,507 nominal training and 2,986,574 test timesteps (>5.06M observations), with per-dataset domain, channel dimension K, sequence lengths, contamination ratio $\pi\%$, anomaly typologies, and partition protocols.

A machine-readable manifest (`reports/stream_manifest.csv`, generated by `scripts/build_stream_manifest.py`) enumerates every stream's identifier, channel dimension, split lengths, held-out anomaly counts, benchmark role, and evaluated seed set as recorded in the released artifacts. All counts in this section, the abstract, and the results tables reconcile against it: 33 primary held-out streams (19 non-GHL at five seeds, 14 GHL at three to five seeds depending on size) and 12 case-study streams.

IV. Methodology: Latent Representation Forecasting and Structured Inductive Priors

This section formalizes the mathematical formulation, latent predictors, and inductive biases of the proposed framework.

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T\times K}$ denote a multivariate stream partitioned into context $x_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and suspect horizon $x_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive baselines (TimesNet, TranAD, Anomaly Transformer) minimize pointwise error $\min_\theta \mathbb{E}\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, which can weight channels with different noise scales unequally; whether this consumes model capacity is not tested here. We instead adopt the JEPA framework [7]: a context encoder E_θ maps $x_{\text{ctx}} \to z_{\text{ctx}} \in \mathbb{R}^D$ and an EMA momentum target encoder E_ϕ maps $x_{\text{tgt}} \to z_{\text{tgt}}$, with $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$, $m_k \in [0.996, 0.9995]$. The objective forecasts target representations in latent space, testing whether structured priors condition prediction residuals for tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

To examine whether incorporating structured dynamical inductive biases stabilizes representation forecasting, we evaluate an unconstrained latent predictor alongside three candidate formulations inspired by physical dynamical principles:

1) Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion): Motivated by Koopman operator theory [33], [34], in which nonlinear trajectories are analyzed via linear operators on observable functions, we parameterize the predictive latent transition via a learned operator $\mathbf{K} \in \mathbb{R}^{D\times D}$: $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent collapse onto a low-rank subspace (dimensional collapse) or to zero (energy collapse), the regularizer decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote the singular values of $\mathbf{K}$, $\lambda_i = \sigma_i^2$ the eigenvalues of $\mathbf{K}\mathbf{K}^T$, and $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_i \lambda_i$ the total energy. For $\mathcal{E}(\mathbf{K}) > 0$, the scale-free spectral distribution over modes is $p_i = \sigma_i^2/\|\mathbf{K}\|_F^2$, $\sum_i p_i = 1$. We omit isotropic diagonal loading from the Gram matrix before normalization — it would give $\mathbf{K} \to \mathbf{0}$ an artificial identity spectrum with maximal entropy, spuriously rewarding collapse. The Von Neumann spectral entropy uses the convention $0\log 0 = 0$:

$$\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i:p_i>0} p_i \log p_i. \tag{1}$$

The *Spectral Entropy Deficiency* quantifies the Kullback-Leibler divergence from the uniform modal equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^T$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0. \tag{2}$$

Here, $\Delta\mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), while dimensional collapse to rank 1 yields maximal deficiency $\Delta\mathcal{H} = \log D$.

To penalize energy collapse ($\|\mathbf{K}\|_F \to 0$) and prevent explosive gain, we introduce a non-negative Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r_{\text{safe}}(\mathbf{K}) - 1 - \log r_{\text{safe}}(\mathbf{K}) \geq 0, \tag{3}$$
$$r_{\text{safe}}(\mathbf{K}) = \max\left(\frac{\|\mathbf{K}\|_F^2}{\gamma D}, \epsilon\right),$$

where $\epsilon = 10^{-6}$ provides a numerical floor. For the identity operator $\mathbf{K} = \mathbf{I}$, $\|\mathbf{I}\|_F^2 = D$, yielding $\Phi_{\text{energy}}(\mathbf{I}) = 0$ exactly. We emphasize that the implemented barrier is *floored*: it attains a finite maximum penalty $\epsilon - 1 - \log\epsilon \approx 12.8$ (before β_{norm} weighting) throughout $\|\mathbf{K}\|_F^2 < \epsilon\gamma D$, rather than diverging; divergence to $+\infty$ is a property only of the unfloored idealization $\Phi^*_{\text{energy}} = r - 1 - \log r$. The operator is initialized near identity ($\mathbf{K} \sim \mathbf{I} + \mathcal{N}(\mathbf{0}, 0.01^2\mathbf{I})$), so $r_{\text{safe}} \approx 1$ and training starts with a vanishing energy penalty; since

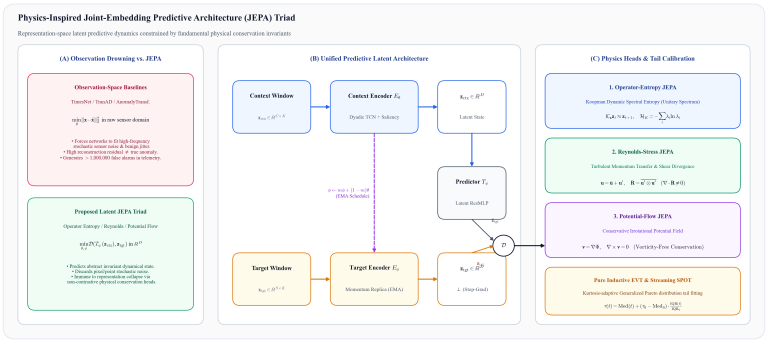


Fig. 1. JEPA telemetry-monitoring architecture: context and suspect windows are encoded and compared in latent space under three candidate physical regularizers, with held-out SPOT calibration.

LayerNorm gives $\mathbb{E}[\|z\|_2^2] \approx D$, the isometric scale $\mathcal{E}_0 = \gamma D$ matches the natural energy of unitary transitions. In sensitivity checks across $\lambda_{\text{ent}} \in \{0.1, 0.5, 1.0\}$ the model converged stably, with 0.5 balancing spectral spread against predictive regression. The composite spectral regularizer is:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K}), \tag{4}$$

with $\beta_{\text{norm}} = 1.0$. The total training objective is:

$$\mathcal{L}_{\text{Op-Entropy}} = \|\mathbf{K}z_{\text{ctx}} - z_{\text{tgt}}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}), \tag{5}$$

with $\lambda_{\text{ent}} = 0.5$ in all experiments. All shared representation components (saliency gating, dyadic frequency shells, LayerNorm) are held fixed across models and controls, isolating the transition-operator loss. We emphasize that $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ acts on the operator matrix, enforcing a non-collapsed operator spectrum; it provides no analytical guarantee against encoder representation collapse, which is mitigated empirically by the EMA target encoder and stop-gradient operations [8], [35].

Proposition 1 (Operator Triviality Invariance)**.** *Let* $\mathcal{L}(\mathbf{K}) = \frac{1}{B}\sum_{b=1}^{B}\|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent} > 0, \beta_{norm} > 0$*, non-trivial target batch* $\frac{1}{B}\sum_{b=1}^{B}\|z_{tgt}^{(b)}\|_2^2 = E_{tgt} > 0$*, and non-degenerate context–target coupling* $\frac{1}{B}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)^T} \neq \mathbf{0}$*. Then (i) the zero operator* $\mathbf{K} = \mathbf{0}$ *is strictly not a local or global minimizer of* $\mathcal{L}(\mathbf{K})$ *under the implemented floored barrier, and (ii) under the unfloored idealization* ($\epsilon \to 0^+$, $\Phi^*_{energy} = r - 1 - \log r$)*, additionally* $\lim_{\|\mathbf{K}\|_F \to 0}\mathcal{L}(\mathbf{K}) = +\infty$.

(Proof in Supplementary Section S-1.)

Remark (Operator Non-Degeneracy vs. Representation Collapse): Proposition 1 concerns only the transition operator: it does not prevent encoder collapse. If E_θ mapped all inputs to a constant $z^{(b)} = \mathbf{c}$, then $\mathbf{K} = \mathbf{I}$ would attain zero prediction error and zero spectral penalty while encoding no information; prevention of that failure mode rests empirically on the stop-gradient momentum dynamics, not on Proposition 1.

2) Reynolds-Stress JEPA (Second-Moment Closure Head): Reynolds decomposition splits a turbulent velocity field into a mean flow and a fluctuation component, where the Reynolds stress tensor captures the second moment of velocity fluctuations. We adapt this second-moment formulation as a latent-space inductive bias; the latent coordinates are learned embeddings, and $\mathbf{\Sigma}$ is not a physical momentum-transport tensor.

A two-layer MLP predicts the conditional mean target latent $\hat{z}_{\text{tgt}}$ from z_{ctx}, and a learned linear projection $\mathbf{P} \in \mathbb{R}^{d_s \times D}$ ($d_s = 16$) applied to the EMA target representation yields the empirical second-moment matrix:

$$\mathbf{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^T + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s \times d_s}. \tag{6}$$

Because $\mathbf{\Sigma}_{\text{obs}}$ is computed from an instantaneous latent vector rather than averaged over the target horizon S, it forms a rank-one outer product with a ridge term enforcing membership in $\mathbb{S}_{++}^{d_s \times d_s}$; statistically it is an auxiliary second-moment projection penalty. The closure head $h_\psi(z_{\text{ctx}})$ outputs a lower-triangular $\mathbf{L} \in \mathbb{R}^{d_s \times d_s}$ with positive diagonal $L_{ii} = \text{softplus}(a_{ii}) + \epsilon$, and the predicted covariance is reconstructed via Cholesky factorization:

$$\mathbf{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^T + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s \times d_s}, \tag{7}$$

confining predictions to $\mathbb{S}_{++}^{d_s \times d_s}$. The training objective is:

$$\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1\ \text{MSE}(\mathbf{\Sigma}_{\text{pred}}, \mathbf{\Sigma}_{\text{obs}}). \tag{8}$$

At inference time, the anomaly score combines the mean latent discrepancy and the fluctuation tensor residual via the coordinate-saliency gate (formalized in Section IV-C). The stress term enters both the training loss and the test score; its downstream contribution is evaluated in Section VII-B and Table I(a).

3) Potential-Flow JEPA (Conservative Field Theory and Flow-Matching Derivation): Under Helmholtz-Hodge decomposition, a smooth field $v \in C^1(\mathbb{R}^D, \mathbb{R}^D)$ splits into a curl-free potential gradient and a solenoidal field, $v = -\nabla\Phi + \nabla \times A$. As dissipative systems relax toward local minima on potential-energy surfaces, we model latent evolution as the dissipative gradient flow $\dot{z}(\tau) = -\nabla_z\Phi_\psi(z(\tau))$ for a scalar potential network Φ_ψ with smooth (SiLU) activations.

Proposition 2 (Absence of Periodic Attractors in Gradient Flows)**.** *Let* $\dot{z} = -\nabla\Phi(z)$ *be an autonomous gradient flow on* $\mathbb{R}^D$ *where* $\Phi \in C^2(\mathbb{R}^D, \mathbb{R})$*. There exist no non-constant periodic orbits.*

(Proof in Supplementary Section S-1.)

a) Continuous Velocity Field and Multi-Subspace Regime Routing: The velocity field is the negative gradient of a parameterized scalar potential, $\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) = -\nabla_{z_t}\Phi_\psi(z_t, t, z_{\text{ctx}}, p_{\text{regime}})$, where Φ_ψ is a three-layer residual MLP (LayerNorm, SiLU) conditioned on state z_t, diffusion

time $t \in [0,1]$ (sinusoidal embedding), context z_{ctx}, and a discrete regime projection p_{regime}; the raw gradient is tangent-projected via $v_\perp = v - \langle u, v\rangle u$, $u = z_t/\|z_t\|_2$, bounding integration curvature.

Discrete operational modes use $K_r = 4$ learnable orthonormal subspace projections $\mathbf{U}_k \in \mathbb{R}^{D\times d}$ (d=8, D=32); the active regime is selected by maximal captured context energy, $k^* = \arg\max_k \|\mathbf{U}_k^\top z_{\text{ctx}}\|_2^2$, $p_{\text{regime}} = \mathbf{U}_{k^*}\mathbf{U}_{k^*}^\top z_{\text{ctx}}$, with an orthogonality separation loss $\mathcal{L}_{\text{subspace}} = \frac{2}{K_r(K_r-1)}\sum_{i<j}\|\mathbf{U}_i^\top \mathbf{U}_j\|_F^2$. Section VI-A includes a single-regime control (K_r=1) with routing disabled.

b) Flow-Matching Objective, Numerical Midpoint Evaluation, and Dynamical Misspecification: Proposition 2 governs the autonomous continuous flow $\dot z = -\nabla\Phi(z)$; the empirical objective instead trains $\hat v$ via conditional flow matching (OT-CFM) along linear paths between noise $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and z_{tgt}: $z_t = (1-t)z_{\text{noise}} + t z_{\text{tgt}}$, $t \sim \mathcal{U}(0,1)$, with target field $v_t^* = z_{\text{tgt}} - z_{\text{noise}}$ and loss

$$\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t, z_{\text{noise}}}\left[\|\hat v(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2\right]. \tag{9}$$

Along the deterministic mean trajectory ($z_{\text{noise}} = \mathbf{0}$), $z_t = t\, z_{\text{tgt}}$ and $v_t^* = z_{\text{tgt}}$ is constant.

At inference, rather than integrating an ODE over $t \in [0,1]$, the detector queries the field at the trajectory midpoint:

$$\mathbf{e}(t) = \hat v(0.5 z_{\text{tgt}},\, 0.5,\, z_{\text{ctx}},\, p_{\text{regime}}) - z_{\text{tgt}}. \tag{10}$$

Since the reference velocity along the mean path is $v_{0.5}^* = z_{\text{tgt}}$, this midpoint query directly measures the discrepancy between the learned potential gradient and the nominal target, matching the dimensionality (D=32) and scale of the unconstrained TS-JEPA residual. The anomaly score applies the saliency gate: $\mathcal{S}(t) = g(\mathbf{e}(t))$.

Dynamical Misspecification on Non-Conservative Telemetry: A curl-free velocity field ($\nabla \times \hat v \equiv \mathbf{0}$) forces circulation along any closed path $\mathcal{C}$ to vanish, $\oint_{\mathcal{C}} \hat v \cdot d\mathbf{r} = 0$. Valid for purely dissipative equilibria, this is dynamically misspecified for the non-conservative cyclic dynamics common in telemetry (motor revolutions, diurnal GECCO consumption cycles, periodic Daphnet gait kinematics): normal circulating orbits are penalized as anomalous. This predicts degraded detection on cyclic telemetry, tested in Section VII-B and Supplementary Section S-4.

C. Anomaly Scoring and Operational Detection Latency

During inference, context windows $x_{\text{ctx}} = \mathbf{X}[t-C : t, :]$ and suspect windows $x_{\text{tgt}} = \mathbf{X}[t : t+S, :]$ are processed at stride 1. For TS-JEPA, Operator-Entropy, and Potential-Flow, the score gates the latent discrepancy through $g(\cdot)$ (Section IV-D1): $\mathcal{S}(t) = g(\hat z_{\text{tgt}}(t) - z_{\text{tgt}}(t))$. For Reynolds-Stress, $\mathcal{S}(t) = g(z_{\text{tgt}} - \hat z_{\text{tgt}}) + 0.5\, g(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}}))$. All shared representation stages are held fixed across variants and ablation controls (Section VI-A).

Operational Distinction: Zero-Lookahead Causal Filtering vs. Buffered Attribution. Two regimes exist, formalized in Figure 2: (1) strictly causal streaming (*trailing* attribution), in which the score emitted at τ is $\mathcal{F}_\tau$-measurable — it uses only observations through window completion; and (2) buffered monitoring (*smear* attribution), in which the identical window score is attributed retrospectively over the suspect region $[t{+}1, t{+}S]$, so an anomaly at t^* can raise discrepancy up to S=64 steps before onset. Under strict unadjusted point-level evaluation (stride 1), alarms during $[t^* - S, t^*]$ score as false alarms: the buffered regime is detection with a fixed S-step buffer, not forecasting. The two regimes are *not* interchangeable evidence; we report them separately throughout (Table II and Supplementary Table S27). Incident analysis (Supplementary Section S-5) also evaluates *Middle* alignment ($t + S/2$); all baselines share identical windowing, horizon, and standardization.

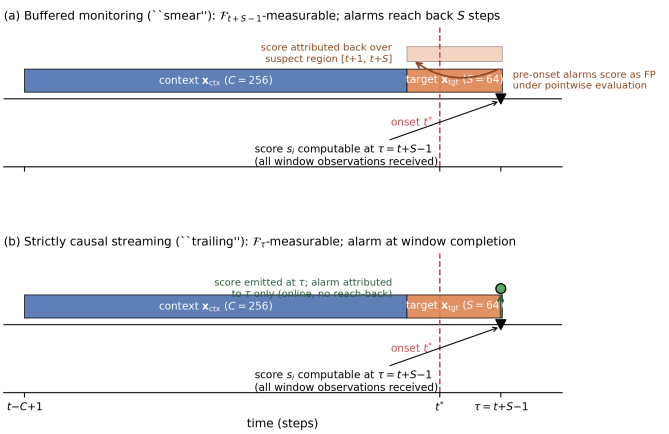


Fig. 2. Score-emission and alarm-attribution timeline. Every score s_i is computable only at window completion $\tau = t{+}S{-}1$. (a) Buffered attribution assigns s_i retrospectively to $[t{+}1, t{+}S]$; (b) causal attribution emits s_i at τ only.

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, all three architectures incorporate three inductive mechanisms:

1) Coordinate Saliency Gating and the Auxiliary Channel-Gate Variant: On high-dimensional streams ($K \geq 25$), faults frequently originate in a localized subsystem while the remaining channels exhibit ambient noise, diluting localized anomalies in isotropic representations. The deployed detector mitigates this in latent space via an anti-drowning coordinate saliency gate on error vectors $\mathbf{e} = z_{\text{tgt}} - z_{\text{pred}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \tag{11}$$

$$w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^{D}\exp(|e_j|/\tau)},$$

amplifying sparse coordinate deviations by up to $\sqrt{D}$ while remaining exactly $\|\mathbf{e}\|_2$ under isotropic errors. This is the discrepancy gate used in every experiment: input channels are encoded unweighted, and the gate acts only on the D-dimensional predictive residual.

a) Auxiliary Input-Channel Gate (Evaluated Variant): We also evaluate a strictly causal *Causal Channel Saliency Gate* as an auxiliary variant (full specification, boundary-condition verification, and the fault-type study in Supplemen-

tary Section S-3): each channel is weighted by the sigmoid of its standardized context-window variance share, $g_k = \sigma((s_k - \bar{s})/(\sigma_s + \epsilon))$, applied symmetrically to context and target encoders so that $\partial g_k / \partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ (zero target leakage, with an explicit identity short-circuit at $K{=}1$). Two boundary conditions apply: quiet faults (flatlines, stuck values) cannot be isolated by variance amplification — the fault-type study confirms they are weakly detected under *every* gating regime (recall 0.02–0.07) — and abrupt regime transitions transiently perturb variance shares until the context window refills.

b) Zero-Leakage Data Provenance and Temporal Isolation: We enforce strict temporal isolation across the training, nominal-calibration, and held-out evaluation partitions (Section V-A): standardization statistics come only from the training partition, EVT tail parameters (u, σ, ξ) only from the calibration partition, and evaluation proceeds chronologically with frozen weights and thresholds. The auxiliary gate's weights depend only on x_{ctx}; the deployed pipeline weights inputs uniformly.

2) Multiscale Frequency Front-End: Anomalies span temporal scales from sensor spikes to thermal drifts; the encoder front-end therefore splits each channel into $J{=}4$ smooth dyadic shells on an $N_{\text{fft}} = 256$ rFFT grid, normalized bin-wise into an exact amplitude partition of unity, $\sum_j \Psi_j[\omega_m] \equiv 1$ (measured deviation $< 1.2 \times 10^{-7}$; construction details in Supplementary Section S-4). Each shell feeds a causal 1D convolution ($k \in \{15, 9, 5, 3\}$ coarse-to-fine), recombined by learned scale routing. Because the shells overlap smoothly at band edges, the front-end is *less* frequency-localized than standard wavelet banks: per-band in-octave spectral mass is $\{0.50, 0.33, 0.20, \approx 1.00\}$ versus $\{0.78, 0.50, 0.55, 0.82\}$ (Haar) and $\{0.86, 0.63, 0.64, 0.87\}$ (DB4) (`reports/lp_filterbank_diagnostics.csv`). We therefore describe it as a Fourier-domain multiscale front-end, not an exact Littlewood–Paley octave decomposition, and measure its empirical contribution by matched ablation (Table Ib).

3) Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities: Binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [15]: excesses over a high threshold u (98th percentile) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ:

$$\tau = u + \frac{\sigma}{\xi}\left[\left(\frac{qN}{N_u}\right)^{-\xi} - 1\right], \tag{12}$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To account for varying tail heaviness across diverse sensor regimes, we modulate the threshold via an empirical heuristic factor governed by the sample excess kurtosis γ_2 of calibration residuals:

$$\tau_{\text{eff}} = \tau \cdot \left[1 + \frac{1}{6}\tanh(\gamma_2)\right]. \tag{13}$$

On heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} rises to suppress volatile-noise false alarms; on light-tailed ones it tightens for sensitivity.

We emphasize that this is an engineering heuristic for pre-asymptotic finite-sample tails, not an asymptotic EVT theorem. In the controlled calibrator ablation (Section VII-D, Table III) it does not improve out-of-sample risk control: on held-out data the multiplier raises TS-JEPA nominal FPR from 5.26% to 17.68% with no statistically distinguishable Point-F1 change (0.1694 vs. 0.1577).

More broadly, EVT does not eliminate threshold-calibration difficulty: under $q = 10^{-3}$ (ideal asymptotic FPR 0.1%), measured held-out nominal FPR settles at 5.3–6.2% for the JEPAs and $\approx 22\%$ for TimesNet/TranAD, owing to finite-sample variance, residual autocorrelation, and non-stationary drift between nominal training and deployment. The calibrated finding of this paper is narrower: latent predictive residuals produce substantially fewer false alarms than observation-space residuals under identical calibration, reflecting lighter tails at the cost of lower causal recall. Thresholds use static EVT on held-out nominal splits; dynamic rolling SPOT is evaluated separately in the operational incident analysis.

V. Experimental Protocol and Evaluation Metrics

The master experiment matrix enumerating all evaluation suites (stream counts, partitions, horizons, calibrators, metrics) is in Supplementary Table S1.

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted using a causal dilated TCN (3 residual blocks, channel depth $F{=}48$, kernel 5, dilations $\{1, 2, 4\}$); global temporal average pooling aggregates the full $C{=}256$ context, with LayerNorm and dropout ($p{=}0.20$). The momentum target encoder follows EMA tracking with cosine schedule $m_{\text{base}}{=}0.996 \rightarrow m_{\text{max}}{=}0.9995$.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\text{min}} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4}, and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences.

a) Held-Out Calibration Protocol: To evaluate out-of-sample threshold calibration without circularity, each stream's nominal telemetry is partitioned into three disjoint chronological blocks: *training* (first 65%; representation weights only), *nominal calibration* (next 17.5%; withheld from training, used solely to fit per-stream GPD tail parameters σ, ξ above $u{=}0.98$ at $q{=}10^{-3}$, with γ_2 estimated strictly within this block), and *nominal evaluation* (final 17.5%, separated by an unobserved 64-step isolation gap; held-out FPR measured on data the threshold never saw). Thresholds then remain fixed during test evaluation to measure static EVT stability under drift; dynamic rolling SPOT is evaluated separately in the operational incident analysis (Supplementary Section S-5).

b) Calibrator Assignment Across Experiments: To avoid ambiguity, we specify the calibrator per experiment. **Primary held-out benchmark** (Tables II, III, Supplementary Table S27; all paired statistics): standard SPOT ($u = p_{98}$, $q = 10^{-3}$, no kurtosis multiplier); headline metrics are the "SPOT" column of Table III, with calibrator variants compared on identical score streams. **Ablations and controlled studies** (Section VI): SPOT with the $\tanh(\gamma_2)/6$ kurtosis multiplier (the released `adaptive_kurtosis` flag). **Buffered-attribution and EVT mechanism studies**: standard SPOT unless the calibrator is the object of comparison. **Incident metrics** (Supplementary Section S-5): rolling SPOT. No experiment mixes calibrators across models within a table.

c) Multi-Seed Replication, Primary Endpoints, and Multiplicity Controls: Principal models on the 19 non-GHL streams are trained across five seeds (42, 123, 456, 7, 2024); the 14 large GHL streams ($> 10^6$ steps) use three seeds (five on the two smallest) for cost, and ablation controls use three. We report held-out nominal FPR, Point-F1, and PR-AUC as distinct outcomes; nonsignificant detection differences are not evidence of parity. The post hoc operating-curve analysis uses only streams with archived score dumps (19 buffered, 18 causal), not the full 33-stream benchmark.

Inference reports stream-macro means, seed SDs, moving-block bootstrap 95% CIs (block L=64), and two-sided Wilcoxon signed-rank tests. The independent unit of analysis is the telemetry stream (n=19 non-GHL at five seeds; n=33 where GHL enters at available coverage): seeds are averaged within stream and repeated windows aggregated to one stream-level score, so nothing is double-counted. Because nonsignificant tests do not establish equivalence, we make no noninferiority claim: paired differences are reported with effect sizes and CIs for readers to judge practical margins. p-values across comparisons are Benjamini–Hochberg FDR-adjusted at $q_{\text{FDR}} = 0.05$; CIs spanning zero or adjusted $p > 0.05$ are treated as inconclusive, not as equivalence. Supplementary Table S28 maps every reported test to its endpoint, cohort, paired-stream count, seed treatment, raw and adjusted p, and effect-size interval.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from external implementation discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline under identical windowing, standardization, and EVT calibration. TimesNet and TranAD are in-house re-implementations of the published architectures (rather than vendor code), configured to published depths and evaluated under the identical optimization budget (AdamW, $\eta = 10^{-3}$, cosine decay, 20 epochs); no method received per-model hyperparameter tuning beyond the shared configuration, and no threshold, early-stopping, or model-selection step accessed held-out anomaly labels. Hardware-specific throughput, batch, warm-up, and per-timestep latency measurements are reported in Supplementary Table S11.

TimesNet [5]: Multi-periodic reconstructive architecture mapping 1D series to 2D variation spaces via FFT period discovery and Inception blocks (648,553 parameters at K=25; 574,529 at K=9).

TranAD [14]: Adversarial reconstructive Transformer with two-stage self-conditioning and sub-series attention (205,746 at K=25; 201,090 at K=9).

TS-JEPA (Unconstrained Control): Standard JEPA predicting z_{tgt} from z_{ctx} via a 2-layer MLP (289,061 parameters at K=25; 218,309 at K=9).

Physics-Inspired Formulations: Operator-Entropy (285,733 at K=25, -1.15% capacity), Reynolds-Stress (295,953, $+2.4\%$), and Potential-Flow (338,117, $+17.0\%$): differences do not stem from parameter scaling.

Linear AR Forecaster (Non-Neural Baseline): Ridge-regularized map from flattened context to target windows, scored by observation-space prediction error — isolating whether latent representation is necessary at all. **Score-bound controls:** a matched-backbone observation-space TCN forecaster with LayerNorm on its bottleneck and flattened output, scored by the same saliency gate; PCA-subspace reconstruction; per-channel max-z; IsolationForest; kNN-distance; observation-space MAD scoring; and the dimensionality-matched PCA→32→LayerNorm gate of Section VI.

C. Multi-Tier Statistical Evaluation Metrics

Critical surveys [28], [36] document severe reporting vulnerabilities in this literature — point-adjustment inflation, channel-averaging imbalances, threshold gaming — so we report metrics across complementary tiers. **Exploratory operating characteristics:** event recall integrated over nominal calibration budgets and, separately, over a fixed exploratory realized-test-FPR interval $[0.10, 0.50]$ (Section VII). For the latter we interpolate within each stream–seed curve only when both bounds are observed, pair models within identical stream–seed cells, average paired differences by stream, and use dataset-level sign permutations and dataset-clustered bootstrap intervals; Benjamini–Hochberg adjustment covers all tested model, mapping, and endpoint combinations. Missing support is reported, not extrapolated. **Secondary endpoints: Dataset-Macro Point-F1** (unweighted mean across datasets, preventing the 14-stream GHL block from dominating); **channel-macro Point-F1**; **PR-AUC** (average precision, ranking quality independent of calibration); **event recall** (fraction of contiguous anomaly segments containing ≥ 1 alarm); **mean detection delay** Δt (onset-to-first-alarm lag over detected events only); and the **trivial all-positive reference** $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at base rate π — models below it underperform a constant alarm. **Pooled micro-F1** exposes flood-prone models; contiguous PA-F1 and test-sweep PA-F1 quantify the residual calibration gap.

D. Evaluation Regimes: Zero-Lookahead Causal vs. Buffered Inference

To keep the buffered-attribution regime separate from future-observation access, we evaluate two regimes: **strictly causal online detection** (at decision time T only $\{X_\tau\}_{\tau \leq T}$ is accessible; scores use trailing attribution; zero lookahead) and **buffered window detection** (S=64 future timesteps are buffered before scoring, capturing full trajectory evolution at a deliberate 64-step latency). Both record Mean Detection Delay, Event Recall within delay budgets ($\Delta t \leq 10, 64$), and false alarms per 1,000 steps ($\text{FAR}_{1\text{k}}$).

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the physics-inspired regularizers benefit detection, Table I(a) reports controlled ablations across all 19 non-GHL streams (principal models at five seeds, controls at three, under the held-out protocol), disabling each physical term on a fixed backbone. Because controls run three seeds against the five-seed principal runs, the paired control means below are computed on the shared three-seed subset and differ slightly from the five-seed Table I(a) rows:

1) **Operator-Entropy vs. $\lambda_{\text{ent}} = 0$:** Disabling the spectral regularizer leaves Point-F1 unchanged (0.241 ± 0.014 vs. 0.244 ± 0.024; difference -0.003, 95% interval $[-0.011, +0.005]$, Wilcoxon $p = 0.62$); PR-AUC stays 0.347–0.348.
2) **Reynolds-Stress vs. $\alpha_{\text{stress}} = 0$:** Disabling the stress loss yields 0.203 ± 0.059 vs. 0.208 ± 0.071 ($+0.005$, $[-0.006, +0.016]$, $p = 0.41$); the null is also insensitive to the covariance estimator (temporal/batch covariances shift F1 by ≤ 0.010 over nine streams).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling regime routing *improves* Point-F1 to 0.129 from 0.096 ($p = 0.048$); both underperform TS-JEPA (0.234, $p = 0.013$).

Across the full 19-stream suite, the physical regularizers produce no statistically detectable improvement over unconstrained latent prediction.

B. Ablation 2: Auxiliary Input-Channel Gating Variant

The auxiliary causal input-channel gate is compared against uniform pooling and non-causal controls under identical TS-JEPA configuration (Supplementary Section S-3, Tables S3, S4). The effect is stream-dependent and macro Point-F1 favors uniform weighting (0.118 vs. 0.077): gating roughly doubles recall on transient faults at $\sim 3\times$ the nominal FPR, while quiet faults remain weakly detected under every regime (recall 0.02–0.07). It is therefore an evaluated variant; the deployed pipeline retains uniform input weighting.

C. Ablation 3: Multiscale Dyadic-Shell Front-End

Replacing the standard front-end with the Fourier-domain dyadic shells plus per-shell causal convolutions and learned routing yields a modest, stream-dependent effect (Table I(b)): macro buffered Point-F1 improves from 0.088 to 0.131 across five multivariate streams, concentrated on MSL and SMD, with a small SMAP regression. Two caveats: the block adds $\approx 1.6 \times 10^5$ parameters, and measured shell overlap means octave isolation is not attained. The front-end is a modest contributor, not the primary driver.

D. Ablation 4: Threshold Calibration Rules on Nominal Splits

The full calibrator comparison is in Section VII-D; two findings bear on the ablations: non-parametric rules match SPOT (validation-max FPR 0.0553 vs. 0.0526 at equal F1), and the kurtosis-adaptive multiplier degrades calibration ($0.0526 \rightarrow 0.1768$ FPR at indistinguishable F1).

E. Controlled Backbone Comparison: Latent Space vs. Observation Space

Does predicting in a contracted latent space condition residuals for EVT thresholding, or is the gain attributable to prediction, compression, the EMA target encoder, or merely the bounded low-dimensional score space? To isolate the mechanism, all 19 non-GHL streams were re-run under the primary held-out protocol (five seeds, identical windows, AdamW budget, and SPOT u=0.98/q=10^{-3} calibration) with five arms sharing the `HybridTCNEncoder` backbone (C=256, S=64, d=32, three dilated TCN layers): **TCN-Obs-Recon** (observation reconstruction), **TCN-Obs-Pred** (observation forecasting), **JEPA-CtxRecon** (latent "prediction" of the EMA-encoded *context* — compression without foresight), **JEPA-NoEMA** (latent prediction with an online hard-copy target), and **TS-JEPA**. Five further controls probe the score-space confound and gate equalization directly: **TCN-ObsPred-LN** (same forecaster backbone, LayerNorm on the 32-dimensional bottleneck and flattened observation forecast, identical saliency gate), **PCA-LinPred-LN**, **Obs-MAD**, **IsoForest**, **KNN-Dist** (training-free observation-space scorers), and **TS-JEPA-NoGate** (identical model, ungated ℓ_2 discrepancy). Supplementary Table S18 reports the per-arm residual diagnostics; the earlier 15-stream multi-SNR study remains in Supplementary Table S2.

Bounded scoring conditions kurtosis, not necessarily threshold transfer: Observation-space objectives produce heavy tails regardless of direction (TCN-Obs-Recon vs. TCN-Obs-Pred: γ_2 8.31/8.24, FPR 0.303/0.310). Latent objectives compress tails $\approx 8\times$ (γ_2 1.0–1.6) and cut FPR $\approx 3\times$ (TS-JEPA -0.230 vs. TCN-Obs-Pred, p=0.031 unadjusted). This could be a mechanical consequence of scoring bounded LayerNorm-ed 32-d vectors rather than learned dynamics. Controls separate the ingredients: **PCA-LinPred-LN** (32-d PCA embeddings, ridge forecast, LayerNorm, identical gated norm — in observation space) reproduces the conditioning and the reduced FPR (γ_2=1.16, FPR 0.052 vs. 0.076), as does **IsoForest** ($\gamma_2 = -0.24$, FPR 0.037). The new **TCN-ObsPred-LN** arm has similarly light trailing tails (γ_2=1.16) but a buffered SPOT FPR of 0.186 and a causal FPR of 0.117, so light kurtosis alone does not preserve threshold transfer across blocks; robust normalization alone (**Obs-MAD**) also floods (0.29).

Conditioning is not detection: At matched nominal budgets PCA-LinPred-LN attains recall 0.665 and IsoForest 0.631 versus 0.774 for TS-JEPA, while removing the saliency gate changes little (0.768). These descriptive budget-point differences cannot isolate an embedding effect because realized FPR differs; the wider matched-FPR check finds no BH-significant pairwise difference and gives the bounded TCN higher buffered recall area on the streams where both curves are defined.

Compression vs. prediction vs. EMA: JEPA-CtxRecon — which never predicts — retains most of the conditioning (-0.203 vs. TCN-Obs-Pred, p=0.055). Removing the EMA target changes little (JEPA-NoEMA: -0.033 vs. TS-JEPA, p=0.17); we cannot conclude equivalence or bound the effect tightly at n=19. PR-AUC differences across the arms are

TABLE I

ABLATION STUDIES. (A) MATCHED-CAPACITY REGULARIZER CONTROLS ON FIXED BACKBONES, 19 NON-GHL STREAMS (POINT-F1 = STREAM-MACRO MEAN ± SEED SD; HELD-OUT FPR ON UNSEEN NOMINAL DATA). (B) DYADIC-SHELL FRONT-END ON/OFF UNDER IDENTICAL TS-JEPA CONFIGURATION. AUXILIARY INPUT-CHANNEL GATING AND FAULT-TYPE RECALL PANELS ARE IN SUPPLEMENTARY SECTION S-3.

(a) Regularizer Controls on Fixed Architectures

Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall
TS-JEPA (Unconstrained)	0.234 ± 0.030	0.323	0.076	0.481
Operator-Entropy JEPA	0.237 ± 0.013	0.323	0.086	0.506
$\lambda_{\text{ent}} = 0$ control	0.244 ± 0.024	0.347	0.083	0.518
Reynolds-Stress JEPA	0.207 ± 0.052	0.307	0.078	0.434
$\alpha_{\text{stress}} = 0$ control	0.203 ± 0.059	0.311	0.066	0.414
Potential-Flow JEPA	0.089 ± 0.036	0.201	0.091	0.303
no-regime control ($K_r = 1$)	0.129 ± 0.061	0.265	0.081	0.392
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490

(b) Dyadic-Shell Front-End On/Off

Stream	Standard Conv	Dyadic Shells	Δ
SMAP `P-3`	**0.294**	0.261	−0.033
MSL `M-1`	0.002	**0.052**	+0.050
SMD `machine-1-2`	0.000	**0.177**	+0.177
Daphnet `S01R01E1`	0.144	**0.165**	+0.021
GECCO `water_quality`	0.000	0.001	+0.001
Macro Mean	0.088	**0.131**	+49.6% rel.

inconclusive ($p \geq 0.16$).

Native Window Horizons: Baselines do not suffer merely from the long W=320 window: at their native horizons tail conditioning *worsens* (Supplementary Table S17). TranAD at W=10 spikes to $\gamma_2 = 448.25$ and TimesNet at W=96 reaches $\gamma_2 = 98.14$ with 44.09% empirical FPR, while TS-JEPA retains $\gamma_2 = 3.14$: this limited horizon check does not validate the in-house re-implementations or exclude other baseline configuration effects.

VII. RESULTS AND DISCUSSION

A. Benchmark Performance on Continuous Telemetry

Table II reports the comparison across 33 continuous telemetry streams under the held-out calibration protocol of Section V-A (SPOT, $u = 0.98, q = 10^{-3}$). Primary findings:

False Alarm Suppression on Held-Out Telemetry: On the 19 non-GHL streams, latent predictors achieve held-out nominal FPR 0.076–0.086 versus 0.300 (TimesNet) and 0.307 (TranAD). Across all 33 streams the paired reduction is significant: TS-JEPA −0.153 vs. TimesNet (CI $[-0.247, -0.071]$, p=0.002) and −0.151 vs. TranAD (p=0.003); Operator-Entropy shows the same effect (−0.144/−0.142, p=0.006). The mechanism is isolated under the same-backbone control (Section VI-E).

Point-F1 and Continuous Ranking (PR-AUC) Uncertainty: False-alarm suppression does not translate into statistically distinguishable detection gains. On the non-GHL subset TS-JEPA attains 0.234 Point-F1 / 0.323 PR-AUC versus 0.197/0.329 (TimesNet) and 0.130/0.287 (TranAD). Across all 33 streams the paired differences are inconclusive: vs. TimesNet +0.012 F1 (CI $[-0.053, +0.075]$, p=0.54), −0.000 PR-AUC (p=0.39); vs. TranAD +0.051 F1 (p=0.88), +0.023 PR-AUC (p=0.16).

Event Recall and Detection Conservatism: Lower FPR accompanies lower event recall: on non-GHL streams TimesNet captures 58.3% of events and TranAD 49.0%, versus 48.1% (TS-JEPA), 43.4% (Reynolds), 30.3% (Potential-Flow). Subtle onsets that do not shift dominant latent coordinates are filtered with the noise.

Detection Delay Trade-Offs (Δt): Mean detection delay (Table II) varies widely: TimesNet responds fastest (Δt=48 steps), TS-JEPA averages 80, and the physics variants are slower still (133–192); TranAD is slowest (234). Part of this gap is a detected-set artifact — delays are averaged over different event sets per model. On the intersection of events both models detect at q^*=0.05, delays are 59 versus 57 steps (TS-JEPA vs. TimesNet) and 60 versus 51 (vs. TCN-ObsPred): the raw gap mostly reflects which events each model catches, not intrinsic latency.

Necessity of Nonlinear Representation Learning (Linear AR Control): A ridge-regularized linear AR forecaster under identical calibration attains FPR 0.006 but collapses to Point-F1 0.057: observation-space linear dynamics cannot separate faults from background variance. Two training-free references sharpen the point (Supplementary Table S22): a PCA-subspace reconstructor matches TimesNet almost exactly (0.311/0.198/0.331 vs. 0.300/0.197/0.329 FPR/F1/PR-AUC), while a per-channel max-z scorer attains the best ranking (0.374) at the worst FPR (0.339): ranking quality and threshold conditioning are independent properties.

Domain-Specific Telemetry Dynamics: Relative performance depends on the physical regime (per-dataset breakdown in Supplementary Tables S15–S16): TimesNet leads on Daphnet periodic kinematics (0.245 vs. 0.155 F1); latent prediction leads on spacecraft telemetry (SMAP: OpEntropy 0.591 PR-AUC; MSL: 0.445 vs. 0.159). On GHL and Metro all models struggle (0.000–0.022 F1).

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

Does adding physics-inspired regularizers (Koopman spectral entropy, Reynolds-stress closure, curl-free potential flow) improve detection over unconstrained `TS-JEPA`? The 33-stream data do not support an aggregate gain:

1) **Operator-Entropy vs. TS-JEPA:** paired difference +0.002 Point-F1 (CI $[-0.025, +0.029]$, p=0.71), +0.002 PR-AUC (p=0.86), +0.009 FPR (p=0.75); leads on 9 of 33 streams.
2) **Reynolds-Stress vs. TS-JEPA:** −0.015 Point-F1 ($[-0.042, +0.007]$, p=0.61), −0.007 PR-AUC (p=0.13); leads on 11 of 33.
3) **Potential-Flow vs. TS-JEPA:** Point-F1 differs by −0.081 under SPOT (p=0.033 unadjusted). Its higher event-recall area over nominal causal budgets is descriptive; without matched realized-FPR support and multiplicity-adjusted evidence, neither an intrinsically stronger detector nor a purely operating-point explanation is established.

These findings align with the matched-capacity controls in Table I(a): the physical regularizers provide no aggregate

TABLE II
HELD-OUT CALIBRATION BENCHMARK, 33 TELEMETRY STREAMS (7 DOMAINS). THRESHOLDS: SPOT (q=10^{-3}) ON HELD-OUT NOMINAL SPLITS. METRICS: HELD-OUT NOMINAL FPR (ON DATA THE THRESHOLD NEVER SAW), UNADJUSTED POINT-F1, PR-AUC, EVENT RECALL, MEAN DETECTION DELAY Δt. "NON-GHL": STREAM-MACRO OVER 19 STREAMS AT FIVE SEEDS; "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS; "DATASET-MACRO": UNWEIGHTED MEAN OF THE 7 DATASET MEANS. BEST IN **BOLD**, SECOND UNDERLINED.

	Non-GHL Streams (19 Streams, 5 Seeds)					Summary	Gasoil Heating Loop (14 Streams)	
Model Architecture	**Held-Out FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**Event Recall ↑**	**Mean Delay (Δt) ↓**	**DS-Macro F1 ↑**	**Held-Out FPR ↓**	**Point-F1 ↑**
Latent-Space Predictive Architectures (JEPAs)								
TS-JEPA (Unconstrained Control)	0.076	0.234	0.323	0.481	80.3	**0.148**	0.005	0.000
Operator-Entropy JEPA	0.086	**0.237**	0.323	0.506	192.3	0.130	0.012	0.001
Reynolds-Stress JEPA	0.078	0.207	0.307	0.434	132.8	0.122	**0.003**	0.002
Potential-Flow JEPA	0.091	0.089	0.201	0.303	146.9	0.065	0.011	0.006
Observation-Space Reconstructive Baselines								
TimesNet (ICLR 2023)	0.300	0.197	**0.329**	**0.583**	48.5	0.100	0.061	**0.022**
TranAD (VLDB 2022)	0.307	0.130	0.287	0.490	234.4	0.069	0.046	0.020
Non-Neural Linear Baseline								
Linear AR Forecaster	**0.006**	0.057	0.262	0.478	**0.0**	0.038	**0.002**	0.000

detection advantage over unconstrained latent prediction. The dynamical diagnostics on *why* (operator rank preservation without detection gain, covariance-shift insensitivity, and decoupling resolution via Lyapunov invariance) are in Supplementary Section S-4.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Event-level incident metrics, latency, and rolling-SPOT case studies are in Supplementary Section S-5: rolling SPOT suppresses false alarms up to $21\times$ on drift-prone streams at a recall cost, and TS-JEPA's causal false-positive rate (4.18%) sits an order of magnitude below reconstructive baselines (20.8–30.1%).

Supplementary Table S27 reports the same benchmark under strictly causal (0-lookahead) streaming. On the 19 non-GHL streams TS-JEPA reduces held-out nominal FPR to 0.0136 versus 0.2427 (TimesNet) and 0.2219 (TranAD), an $\approx 18\times$ suppression; on the 14 GHL streams causal FPR falls below 0.009 for JEPAs, but causal Point-F1 there is near-zero for latent predictors (≤ 0.006 vs. 0.022 TimesNet). Causal Point-F1 drops to 0.1118 ± 0.0412 (TS-JEPA) versus 0.2091 ± 0.0265 (TimesNet), and event recall declines from 58.82% to 46.55%: without a lookahead buffer, latent predictive scoring is an ultra-conservative filter that prevents false alarms at the price of detection sensitivity.

a) Pre-Onset Alarm Attribution: Buffered attribution is strictly retrospective: each window's score is painted over $[t{+}1, t{+}S]$, so no mechanism can generate a lead beyond S=64 steps. The median pre-onset "leads" of 180–204 steps in `reports/early_warning_leadtime.csv` are therefore unrelated nuisance alarms at the underlying false-positive rate or gradual precursor drift, not prediction: TimesNet emits 206 such alarms per 1,000 nominal steps versus 24 for TS-JEPA, a difference consistent with the alarm-rate gap alone. We do not claim early-warning capability.

b) Detection Quality at Matched Operating Points: We fit empirical-quantile thresholds on a common *nominal* budget grid $q \in [10^{-4}, 0.9]$ and evaluate event recall and realized FPR on test. The partial-AUC values 0.742 (TS-JEPA) and 0.618 (TimesNet) integrate recall over *nominal budgets up to* 0.10; they are not matched-realized-FPR areas. At q^*=0.05, causal event recall is 0.774 versus 0.618, but realized FPR differs (0.180 versus 0.262). Pooling scores by budget before integrating a realized-FPR curve (Fig. 3) can conceal stream-specific support and must not be read as a paired detector comparison.

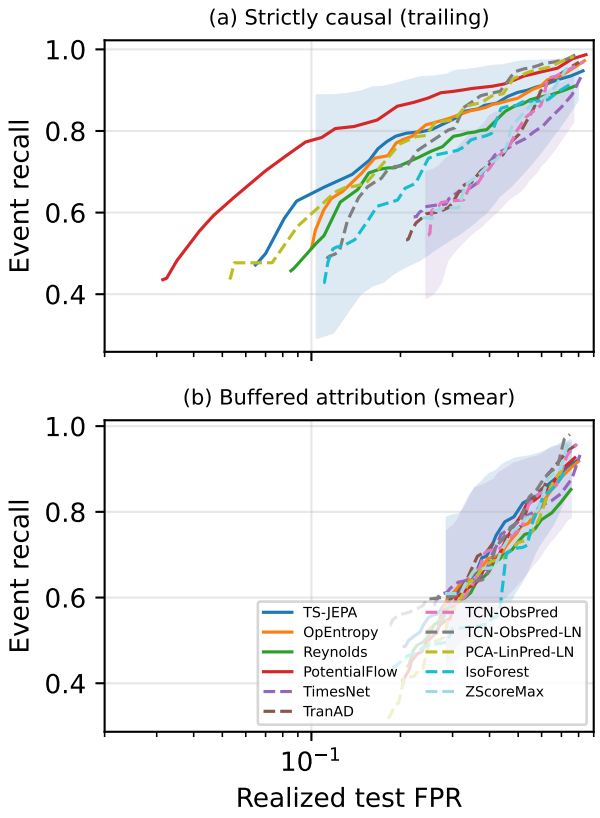


Fig. 3. Pooled event recall versus realized test FPR at common nominal calibration budgets: (a) causal trailing; (b) buffered smear. Bands resample datasets; pooled curves are descriptive, not paired estimates at matched FPR.

For an exploratory support-aware check (interval chosen after inspecting curve coverage), we interpolate each stream–seed curve over the common realized-FPR interval $[0.10, 0.50]$ without extrapolation (`reports/oc_realized_fpr_pairs.csv`). The causal TS-JEPA–TimesNet difference is $+0.061$ (95% dataset-bootstrap CI $[-0.153, 0.329]$; dataset-sign-permutation p=0.625; 20 paired seed cells from four streams). The

TABLE III
CALIBRATOR COMPARISON ON IDENTICAL SCORE STREAMS (33 STREAMS, BUFFERED INFERENCE, UP TO FIVE SEEDS). DESIGN RISK q=10^{-3}=0.1%; THE CI-EXCLUSION COLUMN SHOWS AUTOMATED CALIBRATION SYSTEMATICALLY OVERSHOOTS OUT-OF-SAMPLE.

Model Architecture	Calibrator Rule	Held-Out Nominal FPR	CI Excludes $q = 10^{-3}$	Point-F1
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	**0.0526**	61%	0.1577
	SPOT + Kurtosis ($\beta = 0.167$)	0.1768	82%	**0.1694**
	Empirical 99.5th Percentile	0.0682	55%	0.1598
	Validation Maximum	0.0553	65%	0.1591
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	**0.0617**	65%	0.1599
	SPOT + Kurtosis ($\beta = 0.167$)	0.1229	76%	0.1441
	Empirical 99.5th Percentile	0.0758	62%	**0.1658**
	Validation Maximum	0.0656	64%	0.1627
TimesNet	Standard SPOT ($q = 10^{-3}$)	**0.2217**	86%	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.2281	82%	0.1352
	Empirical 99.5th Percentile	0.2338	87%	**0.1413**
	Validation Maximum	0.2263	86%	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	**0.2217**	82%	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.2453	84%	0.0940
	Empirical 99.5th Percentile	0.2343	84%	**0.0962**
	Validation Maximum	0.2270	84%	0.0952

causal TS-JEPA–PCA-LinPred-LN difference is $+0.121$ (CI $[0.015, 0.310]$, p=0.188; 11 streams). The new bounded observation-space TCN is near TS-JEPA under causal attribution on nine supported streams (-0.018, CI $[-0.055, 0.003]$) and has higher buffered recall area on five supported streams (-0.211, CI $[-0.349, -0.105]$); neither contrast survives family-wide BH adjustment. Thus the nominal-budget ordering and pooled curves do not establish a causal-regime advantage, and boundedness alone does not determine either tail transfer or matched-FPR detection.

D. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does automated extreme-value thresholding adhere to its design risk q out-of-sample? Table III compares calibrators on identical score streams across all 33 streams under buffered inference. The calibrator evaluated here is the *static* peaks-over-threshold step of SPOT — a single threshold fit once on the calibration block; the streaming variant that re-fits on new peaks is evaluated in Supplementary Section S-5. Four realities emerge:

1) **Systematic design-risk overshoot:** the static POT threshold (u=0.98, q=10^{-3}) never achieves its design risk out-of-sample — held-out nominal FPR averages 5.26% (TS-JEPA), 6.17% (OpEntropy), 22.17% (TimesNet), 22.17% (TranAD), a 53–222× overshoot.
2) **High failure rate:** 95% bootstrap CIs exclude q in 61–86% of runs; this protocol does not establish whether drift or scarce effective tail samples dominate the failure.
3) **Parity with empirical quantiles at their own risks:** on TS-JEPA, validation-maximum gives FPR 0.0553 / F1 0.1591 and empirical p99.5 gives 0.0682 / 0.1598 — these encode different nominal risks than q=10^{-3} (not risk-matched); parametric GPD tail modeling confers no benefit at comparable realized FPR.
4) **Kurtosis-adaptive failure (sensitivity arm):** the heuristic multiplier $1 + \tanh(\gamma_2)/6$ inflates TS-JEPA FPR from 0.0526 to 0.1768 (CI-exclusion 61% → 82%) with indistinguishable F1 (0.1694 vs. 0.1577). Note the direction: calibration-block excess kurtosis is non-positive in ∼40% of cells (median 0.23), so the multiplier *lowers* the threshold there — it is an ad-hoc sensitivity arm, not an EVT method, and its failure carries no implication about GPD tail modeling.

To separate the causes of calibration failure, we recorded per-stream tail-fit diagnostics on a nine-stream non-GHL subset (`reports/evt_tail_diagnostics.csv`). Tail-sample scarcity is genuine — the median exceedance count above $u = p_{98}$ is only 21 (range 16–70), and the Grimshaw estimator returns the degenerate shape $\xi = -0.9$ in 33% of TS-JEPA and 52% of TimesNet fits — yet goodness-of-fit is not the dominant failure: KS tests reject in only $6/27$ (TS-JEPA) and $8/27$ (TimesNet) cells, while realized held-out FPR varies from 0.0 to 0.67 across seeds on the *same* stream under passing fits. These observations are compatible with both score drift and tail-sample uncertainty (decomposition in Supplementary Section S-6).

This attribution is quantified stream-level across all dumped cells: the KS distance between calibration and test nominal scores predicts realized SPOT FPR (Spearman $\rho = +0.38$), while calibration-block excess kurtosis predicts *lower* realized FPR ($\rho = -0.32$): the pattern is consistent with a more conservative tail fit, but neither correlation identifies the cause of calibration failure. A controlled-drift study (Supplementary Table S21) sharpens this: runs declustering leaves only 5–9 *effective* exceedances per cell (of a median 21 raw), and rolling calibrators control risk to 1–7% under step, variance, and autocorrelation shifts but fail under continuous ramps (0.33–0.39 FPR): adaptation absorbs regime jumps, not within-block drift.

Computational profiling is in Supplementary Table S11: JEPAs use 183.8–185.3 MB peak memory at 0.60–0.84 ms per-window inference.

VIII. CONCLUSION AND FUTURE WORK

We compared score conditioning, detection, and static threshold calibration on 33 telemetry streams from seven datasets under a held-out protocol. Three conclusions follow.

First, bounded observation-space controls reproduce much of the apparent latent-score tail conditioning, while the bounded TCN shows that light tails do not guarantee threshold transfer. Causal event-recall areas over nominal budgets favor JEPA on average, but paired comparisons at matched realized FPR have sparse support and do not establish a detector advantage; buffered results remain separate. Point-F1 and PR-AUC differences are imprecise, not evidence of equivalence.

Second, the tested physics-inspired variants show no reliable aggregate benefit. Neither the unadjusted Potential-Flow F1 contrast nor its favorable nominal-budget curve establishes a mechanism; explanations based on curl-free dynamics or operating points remain conjectural.

Third, static POT thresholds overshoot their design risk under the short chronological calibration protocol (5.3–6.2% for JEPAs, ≈ 22% for reconstructive baselines at q=10^{-3}). Observational drift diagnostics and scarce tail exceedances do not resolve their relative contributions.

Scope. All evaluations are *within-dataset*: models are trained, calibrated, and tested on disjoint chronological partitions of the same stream. We claim nothing about cross-dataset

transfer or unseen fault types (cf. Section S-3.2). Extensions include solenoidal components via Helmholtz decomposition, memory-bounded online drift adaptation, accelerated streaming inference kernels, and integer quantization for edge deployment.

Code and Data Availability

All benchmark extraction scripts, preprocessing pipelines, checkpoints, training implementations, evaluation runners (scripts/run_multiseed_heldout_benchmark.py, scripts/run_protocol_ablation.py), diagnostic artifacts, and logs are released at https://anonymous.4open.science/r/NCAD_CS-Reproducibility-8F21; public datasets are accessible via their referenced repositories.

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